

Entanglement and Bell's inequality

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Abstract

In this experiment polarization entangled photons are created by spontaneous parametric down conversion (SPDC) of a laser beam passing through a BBO crystal. Using polarizers and synchronized photon detectors we measure coincidence rates of the photon beams in different polarization bases. We start by measuring the visibility in the HV and DA basis, leading to $V_{HV} = 0.9795(19)$ and $V_{AD} = 0.8071(19)$. The lower visibility in the AD basis indicates that photons in both $|\psi_{-}\rangle$ and $|\psi_{+}\rangle$ states are created with a $|\psi_{+}\rangle$ abundance of $9.65(10)$ %. Afterwards we measure coincidence rates for various sets of angles, which we use to compute the Bell parameter $S = 2.395(6)$. This lies below the theoretical prediction of $S_{\max} = 2\sqrt{2}$ but still violates Bell's inequality.

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0 Introduction

Quantum mechanics, with its counterintuitive and almost surreal predictions, continues to challenge our fundamental understanding of nature. Among its most enigmatic phenomena is quantum entanglement, a state in which particles become so deeply interconnected that the properties of one instantly influence the other, even with no information exchange between the particles. This phenomenon seems to defy classical intuitions about the separability of objects in space. The remarkable implications of entanglement reach beyond theoretical curiosities; they call into question the very nature of reality, as they suggest an inherent non-locality in the fabric of the universe.

A groundbreaking development in this area came with John Bell's formulation of Bell's inequality, a theorem that mathematically expresses limits on correlations between particles predicted by classical physics. Violations of Bell's inequality provide evidence for entanglement and the fundamental departure of quantum mechanics from any classical understanding.

In this experiment, polarization-entangled photons are created from a laser beam by spontaneous parametric down conversion. The polarization of pairs of photons is measured. In the first step, the visibility is determined for different bases. The aim of the second part of the experiment is to determine the Bell parameter. Therefore, different combinations of angles for the two polarizers are employed.

The report starts with a brief theoretical overview and then presents the experimental setup and execution. Afterwards, the measurement results are presented.

1 Theory

This section presents the necessary theoretical background for the experiment. The content refers to the script [1] if not stated otherwise.

1.1 Entanglement and Bell states

An important quantum mechanical effect is entanglement. It describes non-local correlations of particles. In our experiment, we focus on the polarization of photons, which can be described as two-level systems. The horizontal and vertical polarizations are distinguished, named $|H\rangle$ and $|V\rangle$. An important property of entangled particles is that they cannot be written as a product state of the wave functions of the two separate particles, as they are inherently connected. There are four maximally entangled states, called Bell states. These are

$$|\psi^+\rangle = \frac{1}{\sqrt{2}}(|HV\rangle + |VH\rangle), \quad |\psi^-\rangle = \frac{1}{\sqrt{2}}(|HV\rangle - |VH\rangle), \quad (1)$$

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + |VV\rangle) \text{ and } |\phi^-\rangle = \frac{1}{\sqrt{2}}(|HH\rangle - |VV\rangle). \quad (2)$$

They are formulated in the basis:

$$|H\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |V\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (3)$$

The Bell states can be expressed in any other orthonormal basis (ONB) too. Explicitly, in the experiment, a Diagonal/Antidiagonal basis is used, which is presented as a rotation of the Horizontal/Vertical basis by 45° . The basis vectors in the new basis are defined as

$$|D\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } |A\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (4)$$

A change of basis vectors does never change the entanglement of the polarizations of two photons.

1.2 Bell's inequality

Bell's inequality is a theorem based on the assumption of hidden variables and the principle of locality. Hidden variables mean that the result of measurements is intrinsically known by the particle, and not only determined by the measurement, as quantum mechanics suggest. In quantum mechanics, the result of the measurement is defined in the moment of measurement. However, with hidden variables, it is defined before, but cannot be observed. The principle of locality describes that no communication between particles can occur, if they are space-like separated. Bell's inequality can be formulated in different ways, depending on the specific purpose. A suitable formulation for this experiment is called the CHSH equation and is defined as

$$S = |E(\alpha, \beta) - E(\alpha, \beta')| + |E(\alpha', \beta) - E(\alpha', \beta')| \leq 2, \quad (5)$$

where S is the Bell parameter and E_i are expectation values, depending on the set angle α for the first polarizer, and β for the second. The angles α' and β' refer to the different angles used. Experimentally, the expectation values can be obtained using the measurement of coincidences. A coincidence refers to the event that both detectors register a photon at the same time (implying both photons passed the polarizers). Using different combinations of polarizer angles, the expectation values are determined as

$$E(\alpha, \beta) = \frac{1}{N} [C_{++}(\alpha, \beta) + C_{--}(\alpha, \beta) - C_{+-}(\alpha, \beta) - C_{-+}(\alpha, \beta)], \quad (6)$$

where C_i are the recorded coincidences for polarizer angles α and β . The $+$ and $-$ signs indicate a 90° shift of the angle α and β , respectively. The expectation value is normalized by dividing by the total number of coincidences N . It can be derived that the expectation value depends on the two angles following

$$E(\alpha, \beta) = -\cos[2(\alpha - \beta)]. \quad (7)$$

Using this relation, we can estimate the combinations of angles α , β , α' and β' for which the Bell parameter is maximized. This yields the angles: $\alpha = 0^\circ$, $\beta = 22.5^\circ$, $\alpha' = 45^\circ$ and $\beta' = 67.5^\circ$. A constant offset can be added to any of these angles and will not change the resulting Bell parameter. This means that there are infinite combinations of angles that yield a maximal Bell parameter, only the difference between the angles is relevant.

1.3 Spontaneous parametric down conversion

An effect to create polarization-entangled photons is called Spontaneous Parametric Down Conversion (SPDC). The effect is based on the second-order optical nonlinearity of a crystal, in our case BBO.

A pump photon with a wavelength of 404 nm is converted to a signal and idler photon, each with wavelengths in the infrared. If the conversion takes place, energy and momentum have to be conserved. That means that

$$\hbar\omega_{\text{pump}} = \hbar\omega_{\text{signal}} + \hbar\omega_{\text{idler}}, \quad (8)$$

where ω_i are the wave numbers of the pump photon, the signal and idler photon, respectively. Analogously, momentum conservation can be formulated as

$$\hbar\vec{k}_{\text{pump}} = \hbar\vec{k}_{\text{signal}} + \hbar\vec{k}_{\text{idler}}, \quad (9)$$

using the different wave vectors $\vec{k}_i$. The used BBO crystal mainly transforms the pump photon to two photons of equal energy. In this experiment, the polarizations of the created photons differ by 90° , thus one is polarized horizontally, and the other one vertically. Due to birefringence of the BBO crystal, the differently polarized photons experience a different phase shift. The two photons are emitted in two light cones, graphically shown in Figure 1. The upper cone represents vertically polarized photons, the lower horizontally polarized ones. On the two intersections, the maximally entangled states $|\psi^-\rangle$ and $|\psi^+\rangle$ occur. Mainly, $|\psi^-\rangle$ are created, but also $|\psi^+\rangle$ occur. In the HV basis, the contribution of $|\psi^+\rangle$ does not reduce the visibility. However, if the AD basis is used, the visibility is decreased significantly in comparison to a pure $|\psi^-\rangle$ state. The mixed state that leaves the BBO crystal can be expressed as

$$|\psi_{\text{mixed}}\rangle = \sqrt{F}|\psi^-\rangle + \sqrt{1-F}|\psi^+\rangle, \quad (10)$$

where F is the probability of the occurrence of $|\psi^-\rangle$, and $1-F$ of $|\psi^+\rangle$.

Using the fidelity F , the density matrix of the mixed state can be calculated as

$$\rho = F|\psi^-\rangle\langle\psi^-| + (1-F)|\psi^+\rangle\langle\psi^+| \quad (11)$$

This matrix can be computed in different bases. The chosen basis influences the visibility V , which is determined as $V = \rho_{22} + \rho_{33} - \rho_{11} - \rho_{44}$. In the HV basis the matrix is

$$\rho_{\text{HV}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1/2 & 1/2 - F & 0 \\ 0 & 1/2 - F & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (12)$$

The visibility then takes on a constant value of 1 and does not depend on F since one can not distinguish between $|\psi^+\rangle$ and $|\psi^-\rangle$ states when measuring in the HV basis. However, in the AD

basis the matrix looks different:

$$\rho_{\text{DA}} = \begin{pmatrix} (1-F)/2 & 0 & 0 & -(1-F)/2 \\ 0 & F/2 & -F/2 & 0 \\ 0 & -F/2 & F/2 & 0 \\ -(1-F)/2 & 0 & 0 & (1-F)/2 \end{pmatrix} \quad (13)$$

This yields a maximal visibility of $V_{\text{DA}} = |2F - 1|$. The absolute value was included to prevent negative visibilities which mathematically occur when $F < 0.5$, eg. there are more photons in the $|\psi^+\rangle$ than $|\psi^-\rangle$ state. In that case one could also set $F = 1 - f$ with f being the ration of $|\psi_+\rangle$ states, thereby flipping the sign of the visibility.

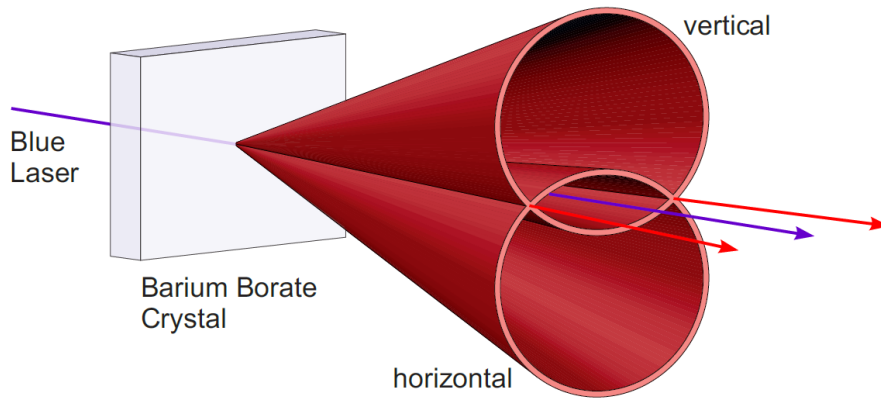


Figure 1: Picture of the BBO crystal splitting up a blue laser beam into photons on two light cones. Taken from [1]

In the setup, two assumptions have to be made. Firstly, we have to assume that the detected photons are a fair sample of all photons, as only a very small fraction of photons is detected. Thus, the detected photons have to represent the whole set of photons. Secondly, no communication between the photons is taking place. Due to geometrical reasons, no space-like separation of the two beam paths is realizable in the setup to really prove, that no communication occurs. To truly test Bell's inequality, space-like separation of the two polarizers, which are the places of the quantum mechanical measurements, has to be achieved.

2 Experimental setup

This section describes the setup that was used for the experiments. A schematic picture of the setup is given in Figure 2. In order to create pump photons, a 404 nm laser is used. The laser beam is directed via two mirrors and a lens, before it enters a BBO crystal. The two beams leave the crystal under some angle and are again redirected by mirrors. Then, each beam transverses a lens, before it passes a Half-wave-plate, which changes the polarization from horizontal to vertical and vice versa. After this waveplate, a BBO crystal of half the length of the first crystal

is placed in each beam path. These are used to compensate the temporal walk-off created in the first crystal, where SPDC occurs. After the crystals, every beam has to pass a polarizer before it arrives at a fiber coupler connected to a single photon detector, where the recorded photon is registered. Although the photons are recorded at the photon detector the quantum mechanical measurement takes place at the polarizer. At this point the polarization state of the photon gets defined.

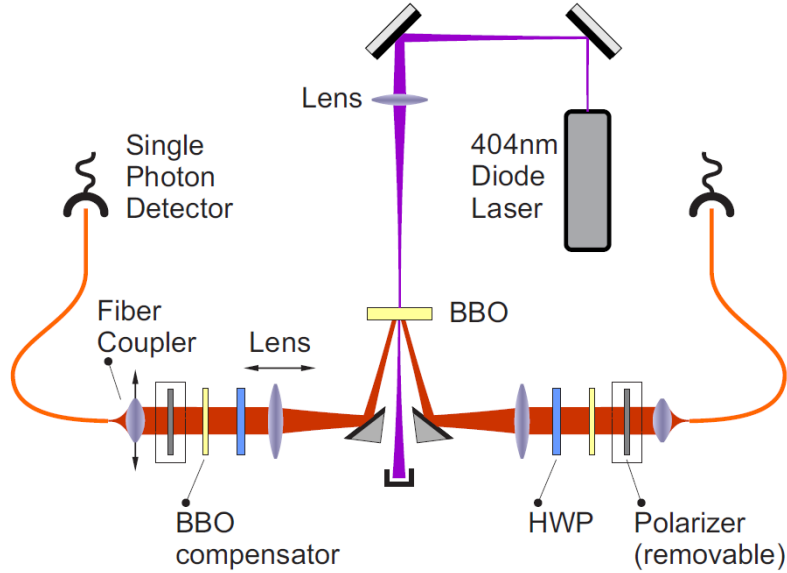


Figure 2: The figure shows the experimental setup schematically. Taken from [1]

Due to birefringence in the first BBO crystal, the vertically and the horizontally polarized photon experience a temporal walk-off. Since the SPDC event does not always occur at the same position in the crystal, the phase shifts also differ from each other. To at least partially compensate this effect, half wave plates and crystals of half the length of the down-converting crystal are used. The waveplates turn the polarization, thus the photon that experienced the lower delay before, now gets a bigger phase shift, and vice versa. The walk-off for different cases can be seen in Figure 3.

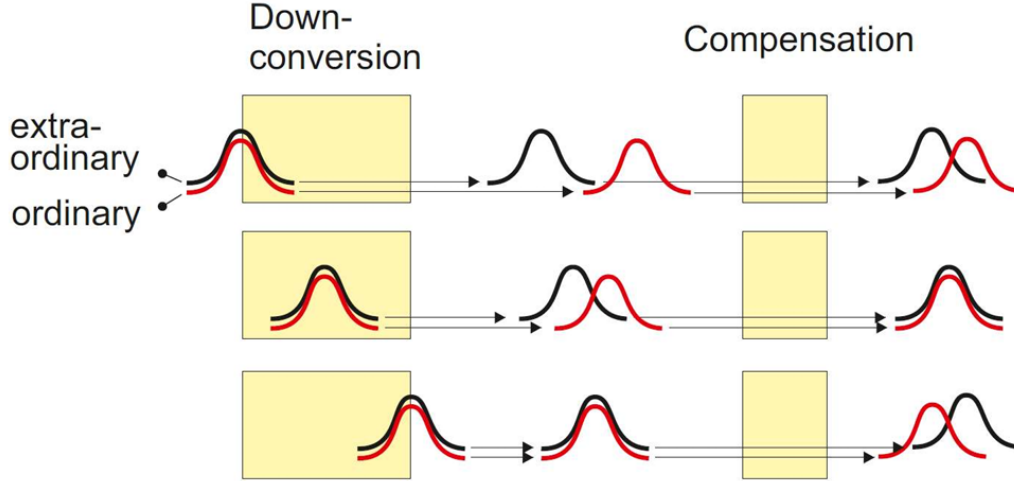


Figure 3: Phase shift of the two photons while transversing the first BBO crystal, where SPDC takes place, and the compensator crystal. Image taken from [1].

The first line shows down conversion at the beginning of the first crystal. Here, a big phase shift is induced, which can be approximately reduced to the half by the compensator. In the second case, the compensator works well and the two photons are nearly in phase afterwards. When the photons are converted at the end of the SPDC crystal, there is nearly no phase shift. The compensator now creates a phase shift. The aim of this is not to undo the phase mismatch that happened in the source crystal (which would require the knowledge of the exact position where the SPDC event happened), but rather to manipulate the photons until they are indistinguishable from each other by virtue of symmetry.

3 Results

We start by analyzing the recorded coincidences for different orientations of polarizers A and B, with A being fixed to a certain angle and letting B sweep angles from 0° to 360° in steps of 5° . For each measurement an integration time of 5 s was chosen. Polarizer A was set to 0° , 90° , as well as 45° and 135° corresponding to polarization measurements in the HV and AD basis respectively. In Figure 4 the recorded data can be seen. In the figure on the top the total photon counts of Polarizer A (B) are depicted as diamonds (dots), with different colors representing different settings of Polarizer A. On the bottom the recorded coincidences are plotted against the sweep angle.

Solid lines represent fitted functions of the form $f(x) = g(x)(\sin^2(\omega x + \phi_0) + y_0)$, with the scaling function $g(x) = a(x - b)^2 + c$. The parameters a, b, c as well as ω, ϕ_0 and y_0 were used for fitting. Our choice of fit function can be motivated the following way: The sinusoidal part describes the physics of interest (eg. an angle dependence of photon coincidence), whilst the parabola tries to correct for artifacts/systematics coming from the experimental setup. Assuming photon coincidences are uncorrelated and happen with a fixed probability (eg. they are poisson distributed), they are directly proportional to the total photon count measured by the two

detectors. Total photon count at detector A stays roughly constant over the whole measurement range for all settings of polarizer B, whilst the photons arriving at detector B show some sine-like dependence, with a noticeable dip around 200° . This can probably be traced back to a small misalignment of the incident laser beam. We chose a parabola to approximate and correct for this unwanted angular dependence as this is rather easy to implement and visually seems to capture the trend. A more rigorous approach would include using fourier analysis to fit a fourier series to the total photon counts. This however exceeds the scope of this report and is therefore left as a suggestion for further investigations.

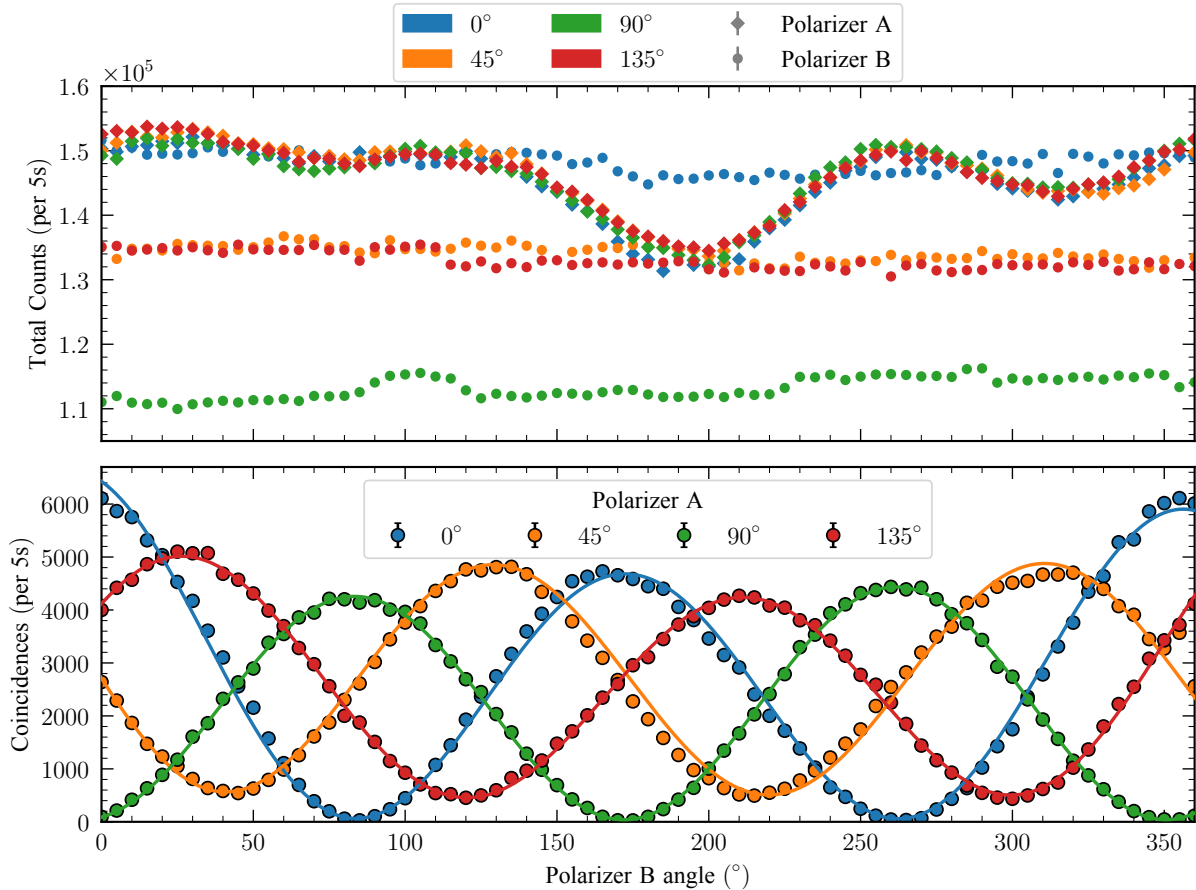


Figure 4: On the top the total photon counts for detector A and B are plotted for four different polarizer A angles (0° , 45° , 90° and 135°) against the varying angle of polarizer B. On the bottom the recorded coincidences are depicted for the same polarizer A angles. A sinusoidal function modulated by a parabola is fitted to each angle sweep and can be seen as solid line. All datapoints have poissonian errorbars which are too small to make out.

We can proceed by using the sinusoidal fits (taking out the parabolic correction) to calculate the visibilities using

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}},$$

with $I_{\max}$ ($I_{\min}$) representing the maximum (minimum) value of the sinusoidal function. Using this we calculate

$$V_{0^\circ} = 0.976(2) \quad V_{45^\circ} = 0.806(3) \quad V_{90^\circ} = 0.976(2) \quad V_{135^\circ} = 0.976(2)$$

From here we can combine the visibilities at 0° and 90° , as well as 45° and 135° to get a mean visibility in the HV and AD basis. We get

$$V_{\text{HV}} = 0.9795(19) \quad \text{and} \quad V_{\text{AD}} = 0.8071(19).$$

The visibility in the HV basis is noticeably larger than in the AD basis. This can be traced back to our entangled photon pairs, which do not purely consist of $|\psi_-\rangle$ states but contain a fraction of $|\psi_+\rangle$ states. In the HV basis both states behave the same way (that is, either both photons are transmitted or reflected), whilst in the AD basis the exact opposite happens. From the measured visibilities we can estimate the $|\psi_+\rangle$ abundance using $1 - F = (1 - V_{\text{AD}})/2 = 0.0965(10) = 9.65(10) \%$.

We investigate the effect of our parabolic envelope by also fitting normal sine functions of the form $f(x) = A \sin^2(\omega x + \phi_0) + y_0$ to the data and calculating visibilities. We get $V_{\text{HV}} = 1.00(2)$ and $V_{\text{AD}} = 0.81(2)$. In both cases the statistical error is larger by an order of magnitude compared to the fits with parabolic envelope. The larger uncertainty is a direct consequence of the pure sine fits which have a smaller fidelity (eg. a larger χ_ν). We conclude that our parabolic envelope successfully decouples some of the systematics from the statistical uncertainty, leading to a more accurate representation of the measured values.

We proceed by performing coincidence measurements with varying polarizer angles to calculate the Bell parameter using Equation (5). More specifically, we measure the coincidences for different combinations of polarizer angles α (for polarizer A) and β (for Polarizer B). The exact angles used can be seen in Table 1. All values in Table 1 were obtained by measuring coincidences in an interval of 60 s with an integration time of 1 s resulting in 60 datapoints for each coincidence measurement, which can be combined to reduce the statistical uncertainty. Additionally, the expectation value can be computed according to Equation (6) and is included in Table 1.

Table 1: All coincidence measurements for different values of polarizer angles α and β . Additionally, the calculated expectation value for each row of coincidence measurements is included in the table.

Polarizer angle (A/B)	α/β C_{++}	$\alpha/\beta + 90^\circ$ C_{+-}	$\alpha + 90^\circ/\beta$ C_{-+}	$\alpha + 90^\circ/\beta + 90^\circ$ C_{--}	$E(\alpha, \beta)$
$\alpha = 0^\circ/\beta = 22.5^\circ$	72(1)	873(4)	435(3)	24.8(6)	-0.862(2)
$\alpha = 0^\circ/\beta = 67.5^\circ$	719(3)	222(2)	104(1)	235(2)	0.491(3)
$\alpha = 45^\circ/\beta = 22.5^\circ$	120(1)	435(3)	356(2)	212(2)	-0.409(4)
$\alpha = 45^\circ/\beta = 67.5^\circ$	135(2)	353(2)	455(3)	47.0(9)	-0.632(3)

Plugging the expectation values for different polarizer angles into Equation (5) we get

$$S = 2.395(6).$$

This value lies significantly above 2, thus violating Bells inequality.

For photons in a pure $|\psi_-\rangle$ state and our choice of angles the Bell parameter reaches a maximal

value of $S_{\max} = 2\sqrt{2}$ (see Equation (7)). Since the value we measured lies below $S_{\max}$ we conclude that our photon pairs were entangled in both $|\psi_{-}\rangle$ and $|\psi_{+}\rangle$ states. We can approximate the fraction F of photons in $|\psi_{-}\rangle$ state by computing $F = S/S_{\max} = 0.847(2)$. This leads us to a visibility of $V = 2F - 1 = 0.693(4)$, which lies significantly below the visibilities measured earlier. Possible explanations for this discrepancy are misalignments in the experimental setup and small inaccuracies when manually setting polarizer angles, all of which decrease the calculated Bell parameter and therefore the derived visibility.

4 Discussion

In this experiment we used entangled photons created by SPDC events in a BBO crystal to determine the Bell parameter. We started by measuring coincidence rates for four different polarization angles to determine the visibility in both HV and AD basis. In the HV basis we expect a visibility of 1 since we can not distinguish between $|\psi_{-}\rangle$ and $|\psi_{+}\rangle$ states. Our measured value $V_{\text{HV}} = 0.9795(19)$ lies below the theoretical prediction, with the statistical uncertainty too small to explain the discrepancy. In the AD basis the predicted visibility depends on the ratio between the two Bell states. Using our measured visibility of $V_{\text{AD}} = 0.8071(19)$ we can estimate that 9.65(10) % of entangled photons are in the $|\psi^{+}\rangle$ state. All mentioned uncertainties only stem from the poissonian nature of counting rates. Systematic errors due to misalignments in the experimental setup were identified but not quantified. We implemented a simple approach to correct for unwanted systematics, reducing the statistical uncertainty by an order of magnitude. Apart from making physical adjustments to the experimental setup, a more sophisticated approach to correcting for unwanted systematics was proposed and can readily be implemented.

We proceeded by measuring coincidences for a set of different polarizer angles to calculate the Bell parameter. The angles $\alpha = 0^{\circ}, \beta = 22.5^{\circ}, \alpha' = 45^{\circ}$ and $\beta' = 67.5^{\circ}$ were chosen to achieve a maximal value of $S_{\max} = 2\sqrt{2}$ for photons entangled in $|\psi_{-}\rangle$ states. We measured $S = 2.395(6)$, which is significantly larger than 2, therefore violating Bells inequality. In order to disprove Theories containing hidden variables, the two detectors have to be spatially separated to be outside of their respective light-cones, which was not investigated but is probably not the case in our setup.

Our measurement is noticeably smaller than the theoretical prediction indicating the presence of $|\psi_{+}\rangle$ photon states. We estimate a $|\psi_{+}\rangle$ abundance of 15.3(2) %, leading to a visibility in the AD basis of $V_{\text{AD}} = 0.693(4)$. This value is well below the visibility measured earlier. Possible explanations for this discrepancy are misalignments in the experimental setup and small inaccuracies which occur when manually setting polarizer angles.

References

- [1] Gregor Weihs, Kaisa Laiho, Harishankar Jayakumar, Christoph Dittel, Robert Keil. *Entanglement and Bell's inequality*. Universität Innsbruck, Innsbruck, AT, WS 2024/25.

Erklärung

Hiermit versichern wir, dass der vorliegende Bericht selbständig verfasst wurde und alle notwendigen Quellen und Referenzen angegeben sind.



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